Algebra Monsoon 2026 Lecture 11 Actions, orders, centres of p-groups

Let G be a group and S a G-set.

We have seen that the set $G\backslash S$ of G-orbits is a partition of S.

Let $\sigma : G\backslash S \to S$ be any section of the canonical surjection $S \to G\backslash S$.

(so that $\sigma_*(G\backslash S)$ is a system of representatives for the partition $G\backslash S$)

Corollary $\#S = \sum_{s \in \sigma_*(G\backslash S)} \#G.s$

Proof This is a special case the following result:

For any mapping $f : X \to Y$, we have $\#X = \sum_{y \in Y} \#f^*(\{y\})$

Let G be a group,

$H \subset G$ a subgroup.

The group H° (the opposite group) acts on the set G by

$$\begin{aligned} H^\circ \times G &\xrightarrow{\varphi} G \\ (h, g) &\longmapsto gh. \end{aligned}$$

Indeed. $\varphi(e, g) = g$ for every $g \in G$.

$$\begin{aligned} \cdot\ \varphi(h_2, \varphi(h_1, g)) &= (gh_1)h_2 \\ &= g(h_1 h_2) \\ &= \varphi(h_1 h_2, g) \end{aligned}$$

for all $h_1, h_2 \in H$ and $g \in G$.

The set of H°-orbits $H^\circ \backslash G$ is denoted G/H. It is the set of all <u>left</u> gH of H. ($g \in G$).

Note: G/H can be viewed as a G-set: $g \cdot xH = gxH$ for $g, x \in G$.

$g \cdot xH = gxH$

Let G be a group, $H \subset G$ a subgroup.

Proposition There is a unique bijection

$$H\backslash G \xrightarrow{\varphi} G/H$$

making the square

$$\begin{array}{ccc} g & G & \xrightarrow{x \mapsto x^{-1}} G & g \\ \downarrow & \downarrow & \downarrow & \downarrow \\ Hg & H\backslash G & \longrightarrow G/H & gH \end{array}$$

commute.

Proof. $Hg_1 = Hg_2$

$\iff g_1 \in Hg_2$

$\iff g_1 = hg_2$ for some $h \in H$

$\iff g_1^{-1} = g_2^{-1} h^{-1}$

$\iff g_1^{-1} \in g_2^{-1} H$

$\iff g_1^{-1} H = g_2^{-1} H,$

so there is a unique mapping φ making the square commute. Similarly, there is a unique mapping $\psi : G/H \longrightarrow H\backslash G$ making the square

$$\begin{array}{ccc} G & \xrightarrow{x \mapsto x^{-1}} & G \\ \downarrow & & \downarrow \\ H\backslash G & \longrightarrow & G/H \end{array}$$



commute.

Juxtaposing the two square both ways shows that $\varphi \circ \psi = Id$ and $\psi \circ \varphi = Id$.

□

Let G be a group,
S a G-set, and $s \in S$.

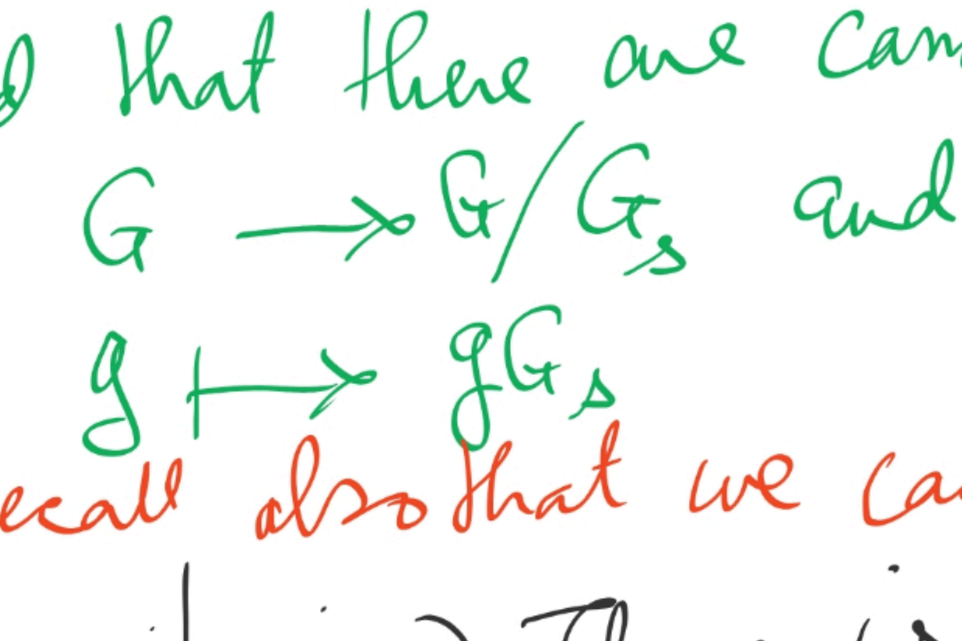
Recall that G_s is the stabiliser of s in G and that there are canonical surjections

$G \to G/G_s$ and $G \xrightarrow{\pi} G.s$

$g \longmapsto gG_s$ $\quad$ $g \longmapsto g.s$

Recall also that we can view G/G_s and $G.s$ as G-sets.

Proposition 1) There is a unique mapping $G/G_s \to G.s$ making the triangle



commute, and it is G-equivariant bijection.

2) $G_{g.s} = g G_s g^{-1}$.

Proof If $g, h \in G$ are such that $gG_s = hG_s$, then there exists a $c \in G_s$ such that $g = hc$ (We then have

$\pi(g) = \pi(hc) = hc.s = h.(c.s)$
$= h.s = \pi(h)$,

so there is a unique mapping φ making the triangle commute.

Let us show that φ is a bijection.

- φ is surjective because π is surjective.
- Suppose $\varphi(gG_s) = \varphi(hG_s)$ for some $g, h \in G$. Then

$$g \cdot s = h \cdot s,$$

so $h^{-1}g \cdot s = s$

so $h^{-1}g \in G_s$

so $g \in hG_s$

so $gG_s = hG_s$

Therefore φ is injective and hence bijective.

Finally, φ is G-equivariant because

$$\begin{aligned}\varphi(g \cdot hG_s) &= \varphi(ghG_s) \\ &= gh \cdot s \\ &= g \cdot (h \cdot s)\end{aligned}$$

$$= g \cdot \varphi(\bar{h} G_s)$$

2) We have

$$h \in G_{g \cdot s}$$

$$\Longleftrightarrow \quad h \cdot (g \cdot s) = g \cdot s$$

$$\Longleftrightarrow \quad g^{-1} h g \cdot s = s$$

$$\Longleftrightarrow \quad g^{-1} h g \in G_s$$

$$\Longleftrightarrow \quad h \in g G_s g^{-1}.$$

□

Corollary The size of every G-orbit divides $\#G$.

Proof: Indeed, the bijection

$$G/G_s \xrightarrow{\sim} G \cdot s$$

shows that the size of $G \cdot s$ equals the index of G_s in G. Now, the index of a subgroup always divides the order of the group, as we have seen.

□

Exercise Let G be a group and S a G-set. If $\#G = 1001$ and $\#S = 6$, then the action of G on S is trivial.

We have to show that $G.s = \{s\}$ for every $s \in S$. Now, $\#G.s$ must divide $1001 = 7.11.13$, and $\#G.s \leq 6$, so $\#G.s = 1$. □

- Let p be a prime. A <u>p-group</u> is a finite group such that whose order is a power of p.

- For a group G and a G-set S, we denote by S^G the set of all <u>fixed points</u>, namely $s \in S$ such that $g.s = s$ for all $g \in G$.

<u>**Proposition**</u> Let p be a prime number, G a p-group and S a G-set.

We have $|S^G| \equiv |S| \pmod p$

Proof S^G can be identified with the set of G-orbits reduced to a point; its complement $C_S(S^G)$ is the union of all G-orbits U such that $|U| > 1$. As $|U|$ also divides $|G|$ (which is a power of p), we must have $|U| \equiv 0 \pmod p$ and consequently $|C_S(S^G)| \equiv 0 \pmod p$. As $|S| = |S^G| + |C_S(S^G)|$, the result follows. □

Proposition If G is a p-group and $|G| > 1$, then $|Z(G)| > 1$.

Proof. Make the group G act on the set G by conjugation; the set of fixed points for this action is $Z(G)$. By the previous proposition, $|Z(G)| \equiv |G| \equiv 0 \pmod p$, so $|Z(G)|$ is a multiple of p and cannot consist

↑ because G is a p-group and $|G|>1$

of $\mathrm{Re}\,\xi$ alone.